

Machine Learning to Predict Climate Change at ClimateWins

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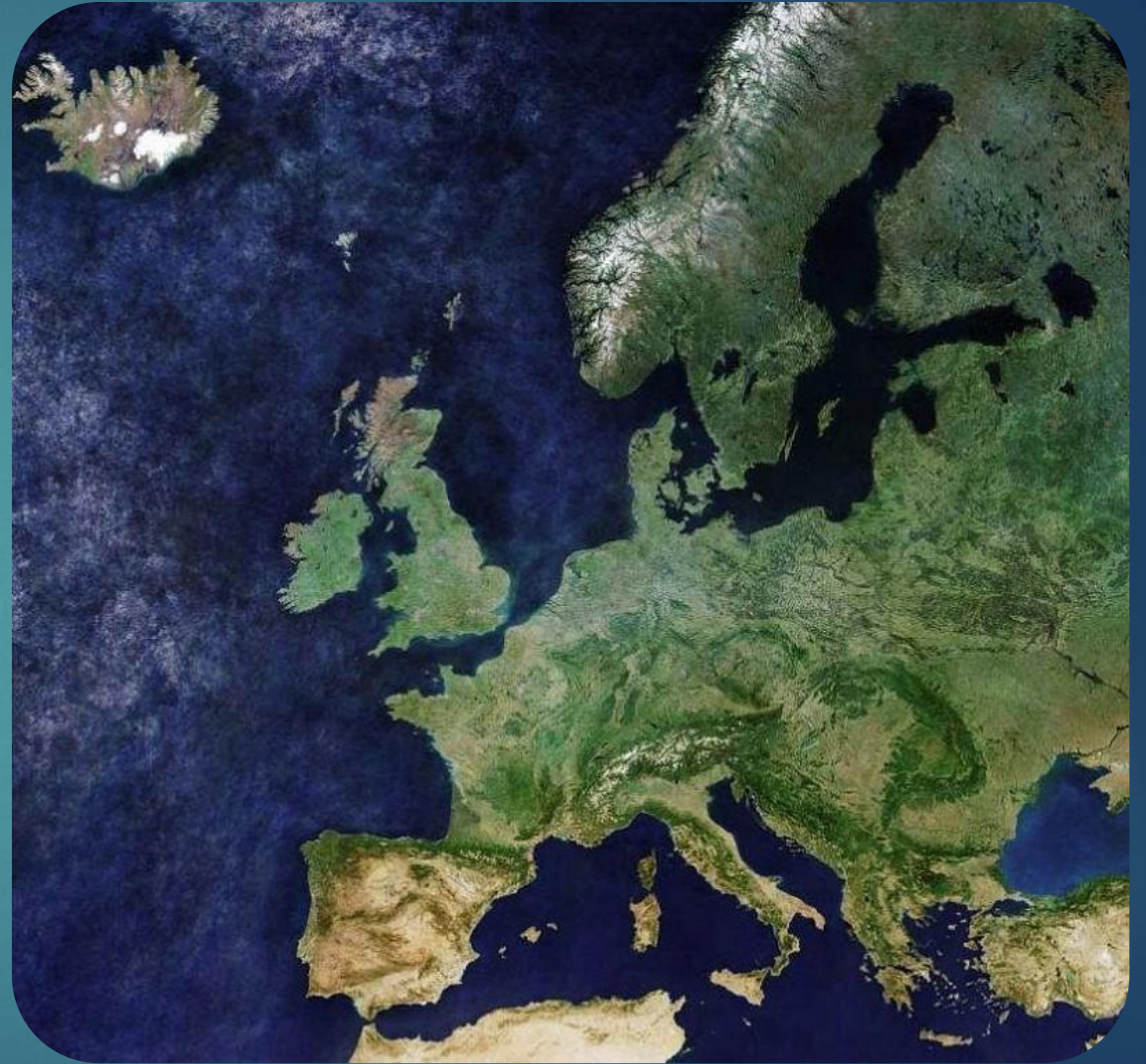


Photo by the Copernicus Sentinel-3A weather satellite of the European Space Agency (ESA)

ClimateWins

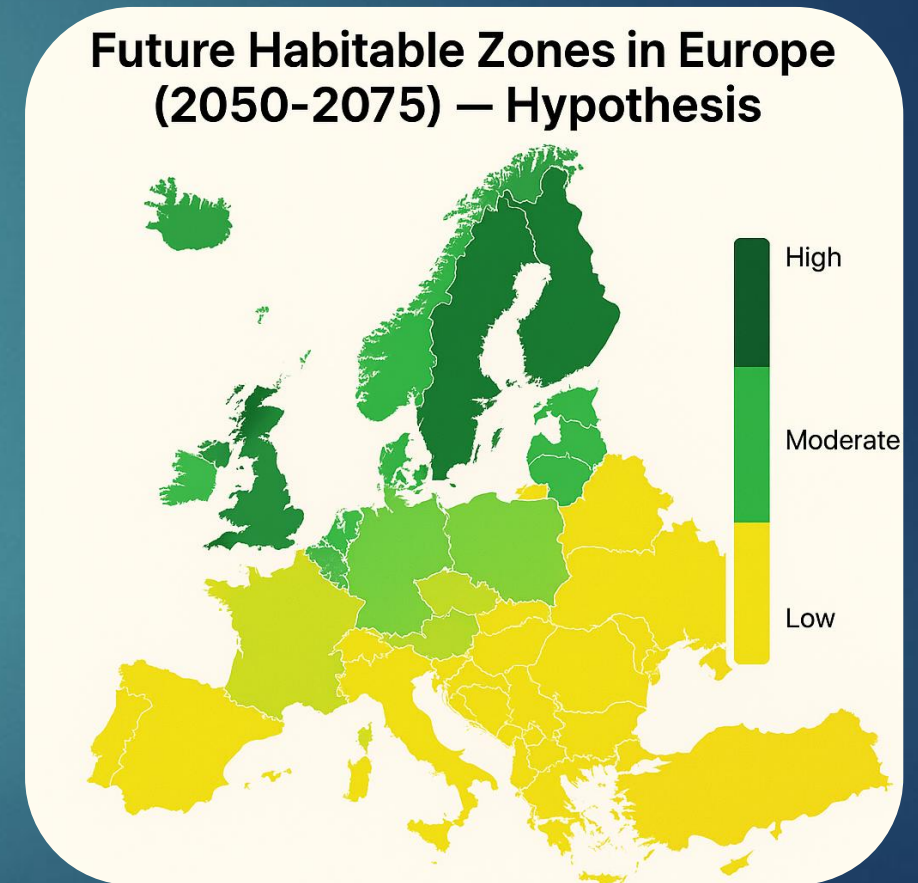
Project: Climate Forecasting

Background

ClimateWins, a non-profit organization, seeks to apply machine learning (ML) to predict the consequences of climate change in Europe and beyond. It is concerned with the acceleration of extreme weather events in recent decades, believing that such fat-tailed risk can be viably modeled and predicted with ML.

Objectives

- Identify European weather patterns outside the regional norm
- Determine if unusual weather patterns are increasing
- Generate future weather scenarios (25-50 years) based on historical trends
- Project the safe habitable zones in Europe over next 25-50 years



AI-rendered hypothetical map by OpenAI's ChatGPT, GPT-4o model

ML models, techniques explored

Algorithms

- KNN (K-Nearest Neighbors)
- DT (Decision Tree)
- RF (Random Forests)
- ANN (Artificial Neural Network)
- CNN (Convolutional Neural Network)
- GAN (Generative Adversarial Network)
- RNN (Recurrent Neural Network)
- LSTM (Long Short-Term Memory as RNN variant)

Analytical techniques

- Gradient descent
- Adaptive moment estimation (Adam)
- Data preprocessing (normalization, reshaping one-hot encoding, subsetting)
- Hyperparameter tuning (grid search, random search, Bayesian optimization)
- Regularization (dropout, L1, L2)
- PCA (Principal Component Analysis)
- Hierarchical clustering (dendrogram)

Experimentation

To develop familiarity with supervised and unsupervised learning ML models (as above), concepts and methods, the majority of ClimateWins-required tasks centered on using historical weather data (1960-2022) to predict a daily 'Pleasant' or 'Unpleasant' weather outcome (label) at each of 15 weather stations in Europe. The "Black Box" labels (answers dataset) in the binary prediction (classification) exercises were of undisclosed origin. However, the accidental discovery of 100% perfect label prediction accuracy on an imbalanced class by the Random Forests algorithm, led to an Occam's Razor explanation: the unknown-origin labels must have been created via deterministic rules and thresholds for key weather features. The Random Forests algorithm excels at reverse-engineering such type of label. Further LSTM-based poking around using a timesteps=1 setting, which achieved 96-99% classification accuracy on five different metrics, revealed the likely use of same-day weather data to create each daily label outcome at a weather station. If the labels were created as believed, then the model-specific evaluations can be limited to a specific use case (classifying deterministic rule and threshold-based labels).

Thought experiments

#1 Detecting and quantifying anomalous weather trends

"If we define what counts as 'anomalous' weather at each station (i.e., based on a local z-score > 2.0 in historical records), can we assume such anomalies increase over time at any of the 15+ European stations? And if so, where and when are they concentrated? Which weather features are most anomaly-prone?"

#2 Forecasting future weather patterns

"Using a Transformer-based time series model like Autoformer, can we simulate plausible daily weather variable trajectories 25–50 years into the future for Europe, while enforcing physical constraints (temperature ranges, energy balance, etc.) to ensure realism?"

#3 Mapping future habitable zones in Europe (future risk assessment)

"What if we assume that the European "comfort zone"—defined as days with temperature 18–27°C, low humidity, low wind speed, and moderate sunshine—shifts northward by 2050–2075? Can we map how the number of "pleasant" days per year changes by weather station and generate a future comfort-zone map?"

Thought experiment #1: *Detecting and quantifying anomalous weather trends*

Hypothesis

The incidence of 'anomalous' weather (station-level weather feature z-score > 2.0 in historical records) has increased across Europe in recent decades, for multiple weather features.

ML model(s) and methods

Isolation Forest and/or z-score for anomaly detection; time series clustering (DBSCAN and/or k-means) over 1-year windows; *Seasonal* Mann-Kendall Trend Test (non-parametric); linear and non-linear (polynomial regression) trend analysis; wavelet analysis; time series decomposition (STL using LOESS); stationarization.

Additional Data needs

Expansion of temporal range of 1960-2022 daily weather data by 40 years. Optional: more features and weather stations, particularly in southern European latitudes. For more robust analysis: elevation, proximity to water bodies, gridded climate reanalysis data (ERA5/ERA20C) to enhance granularity and data uniformity.

Complexity and resources

Low complexity, lower cost.

Thought experiment #2: *Forecasting future weather patterns*

Hypothesis

We can simulate plausible weather scenarios 25-50 years out in time, and impose physical law-based constraints on the models to match reality.

ML model(s) and methods

Hybrid Numerical Weather Prediction (NWP) physics-based model and Transformer-based time series model (Autoformer), where NWP outputs are used as input features for the Autoformer model; synthetic data generation (GANs); time series forecasting (Prophet, ARIMA).

Additional Data needs

Expansion of temporal range of 1960-2022 daily weather data by 40 years. Optional: more features (CO2, solar activity, oceanic indices AMO/ANSO), high resolution gridded datasets (ERA5) to enhance data granularity, uniformity and comprehensiveness.

Complexity and resources

High complexity. NWP is also compute-intensive, high cost. Inaccessible to the non-profit ClimateWins' budget.

Thought experiment #3: *Mapping future habitable zones in Europe*

Hypothesis

The European weather "comfort zone" (*days with 18–27°C temperature, low humidity, low wind speed, and moderate sunshine*) will shift northward by 2050-2075.

ML model(s) and methods

Simplified, reduced-cost version of Experiment #2 for the forecasting part. Random Forests model for the classification part. Clustering, threshold-based classification. GIS (Geographic Information Systems) for integration and visualization. Scenario analysis with synthetic data inputs.

Additional Data needs

Expansion of temporal range of 1960-2022 daily weather data by 40 years. Population density, land use, sea level rise, fire hazard maps, flood risk maps. Climate thresholds for habitability (i.e., max wet-bulb temperature).

Complexity and resources

Moderate.

Conclusion

Recommendation: *Thought Experiment #3 Mapping future habitable zones in Europe*

Rationale: Aligns with public-facing non-profit goals, impactful, visual, action-oriented, lower cost, feasible.

Goal: Project how the frequency and spatial distribution of 'Habitable' days will shift across European regions by 2050-2075.

ML models: Hybrid NWP-Transformer (Autoformer) model, GANs, Prophet, ARIMA or LSTM (Long Short-Term Memory) as backup. Random Forests.

Data: At a minimum, 1925-2025 daily weather data (9+ weather features) at 15+ European weather stations. Habitability thresholds. GIS data. Population density, land use, sea level rise, fire hazard maps, flood risk maps. Simplified, reduced-cost version of Experiment #2 projection data.

Next steps: Acquire data > define habitability thresholds using interdisciplinary literature > run simplified version of Experiment #2 to project weather feature data > fit RF model to simulation data and habitability threshold-based y labels > summarize, evaluate, visualize results > adjust, iterate on models as needed

Thank You

GitHub repository for the Python scripts and datasets used for this project:

[GitHub](#) 

Contact

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Projected % Habitable Days by City (2050–2075)

Assumptions: RCP 2.6 = Low Emissions Scenario; RCP 8.5 = High Emissions Scenario

City	% Habitable Days (RCP 2.6)	% Habitable Days (RCP 8.5)
Reykjavík	54%	71%
Madrid	88%	76%
Oslo	49%	67%
Rome	82%	59%
Berlin	74%	65%
Warsaw	69%	57%
Athens	91%	63%
Paris	77%	68%
Lisbon	85%	79%
Vienna	73%	61%

AI-rendered hypothetical table by OpenAI's ChatGPT, GPT-4o model